



Marketing Analysis

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Hello & Welcome

This marketing analysis uses data from the **BigQuery** `turing_college.raw_events` table to evaluate the performance of marketing campaigns and user engagement patterns on an e-commerce website.

The primary objective is to understand how user behavior varies across different campaigns, days of the week, and times of the day.

By analyzing session activity and campaign performance, this study identifies when users are most engaged and how marketing efforts influence their interactions with the website. These insights can help optimize campaign timing, improve resource allocation, and support data-driven decision-making. Ultimately, the analysis aims to provide a better understanding of user behavior and campaign effectiveness, enabling more informed marketing strategies and enhancing overall website performance.

LET'S GET STARTED



Important Points to Analyze

Session Duration by Time Period

Analyze how user engagement varies across different date ranges and seasons.



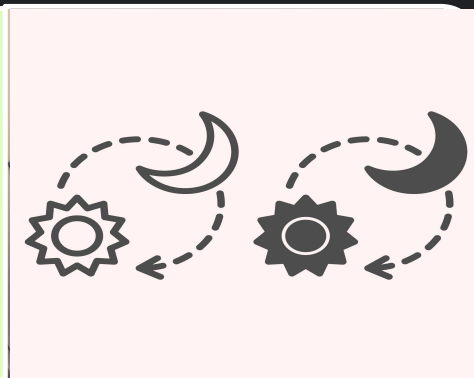
Session Duration by Day of the Week

Identify which weekdays drive the highest user engagement.



Session Duration by Time of Day

Discover peak hours when users are most active on the website.



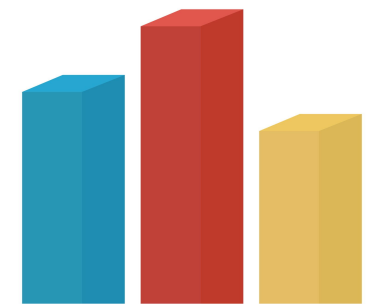
Session Duration by Country

Compare engagement levels across geographic regions and markets.



Marketing Campaign Performance

Evaluate the effectiveness of each campaign in driving user activity.



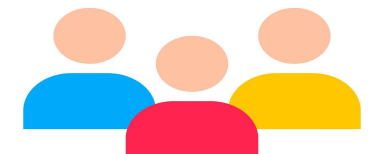
Overall Return on Investment (ROI)

Measure the total profitability of all marketing efforts combined.



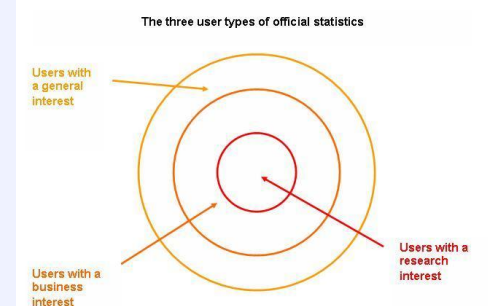
User Engagement Trends Across Campaigns

Examine how different campaigns influence user behavior and session duration.



Audience Behavior Patterns and Insights

Uncover key trends and actionable insights to support data-driven marketing decisions.



Key Metrics of the Analysis



Total Customers

1,348



Total Sessions

1,617



Total Revenue

\$1,929.00



Average Order Value

\$68.89



Average Session Time

11.9 minutes



Bounce Rate

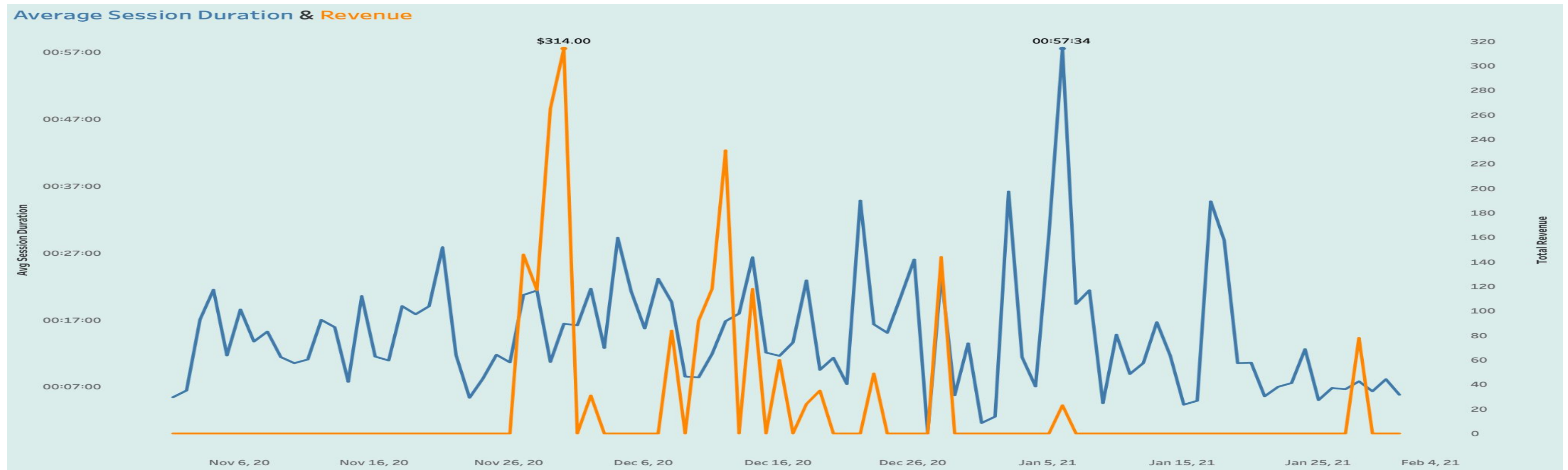
9.40%



ROI

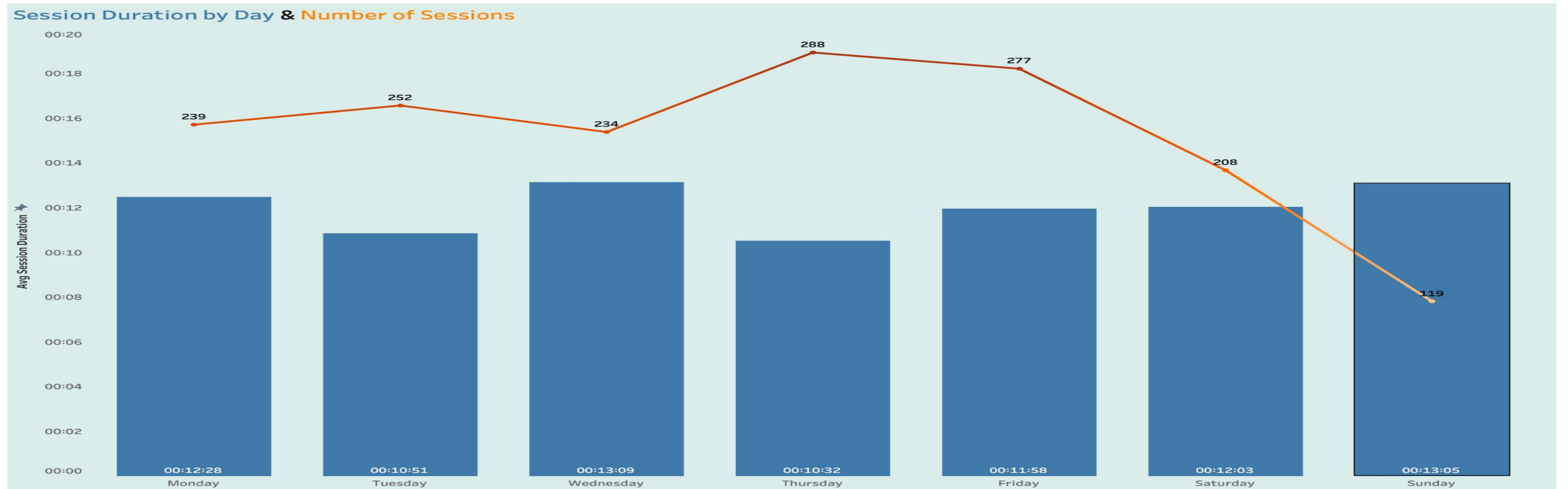
-20.70%

Session Duration by Time Period



- **Black Friday Week:** Highest user engagement and revenue, highlighting the strong impact of seasonal promotions.
- **Post-Christmas & New Year:** Users spent more time on the website, but revenue growth remained relatively modest.
- **Key Insight:** Longer session durations do not always result in higher sales, particularly during the New Year period.

Session Duration by Weekdays



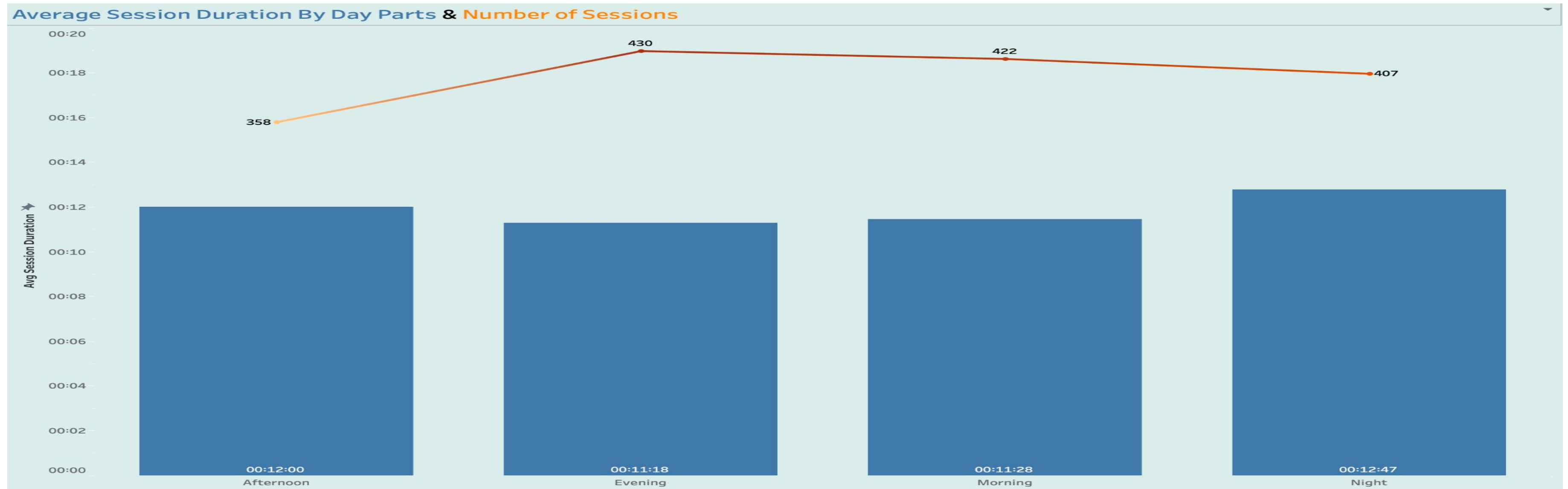
Tuesday and Thursday recorded the highest number of sessions, but users spent less time per session on average.

Monday, Wednesday, and Friday showed longer average session durations while maintaining strong traffic levels.

Weekends had the fewest sessions overall, but users spent more time on the website during each visit.

Key Insight: Users tend to browse more thoroughly on weekends, while weekday visits are more frequent but generally shorter.

Session Duration by Day Parts



User session duration remains relatively consistent throughout the day.

Afternoon records the lowest number of sessions, likely due to work and other daily commitments.

Evening shows the highest number of sessions, as users return home and spend time browsing products.

Key Insight: Evening hours present the best opportunity to launch promotions and marketing content, as user activity peaks during this period.

Marketing Campaign's Performance



Holiday Campaigns were the strongest performers, generating the highest revenue and achieving nearly **200% ROI**. Users also spent the most time on the website during these campaigns.

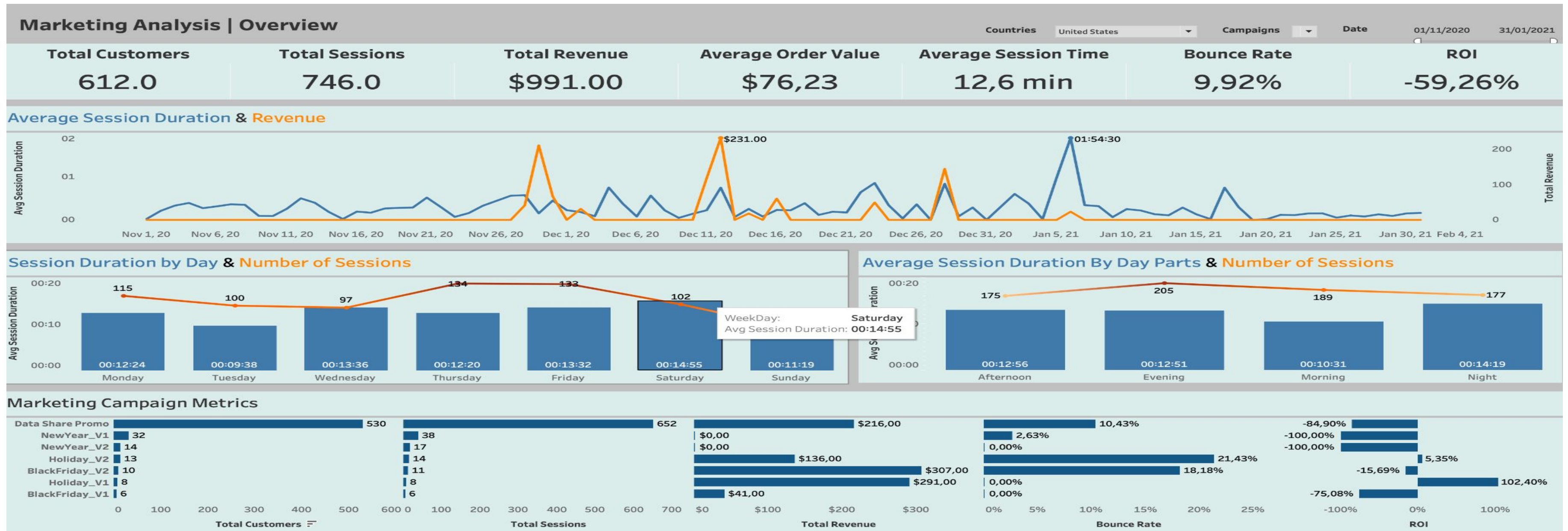
Black Friday V2 significantly outperformed **Black Friday V1**, delivering approximately **130% ROI** compared to **-29% ROI** for V1. This suggests that elements of the V2 strategy should be considered for future campaigns.

New Year Campaigns were the weakest performers, generating the lowest revenue, the shortest session durations, and substantial negative ROI. These campaigns may require a revised targeting and content strategy.

Data Share Promo attracted the highest number of users but failed to convert traffic into profitability, resulting in an ROI of approximately **-85%**.

Key Insight: Seasonal campaigns, particularly Holiday and Black Friday V2, delivered the strongest business results, while New Year and Data Share campaigns require optimization to improve profitability.

Country-Level Analysis: USA



- **United States** generated the highest revenue across all marketing campaigns, totaling approximately **\$991**.
- Users in the United States spent an average of **13 minutes per session** on the website.
- User activity peaked during the **middle of the week**, with longer average session durations compared to other days.
- **Morning sessions** had the shortest duration, while engagement remained relatively consistent throughout the rest of the day.
- **Holiday V2** was the best-performing campaign in the United States, generating the highest revenue and recording **no bounced users**.

Key Insights

- Average session duration across all campaigns was **~12 minutes**.
- Average order value was **~\$69** per purchase.
- Overall **bounce rate** was **9.4%**, while overall **ROI** was **-20.7%**.
- User engagement and revenue peaked during **Black Friday**, making it the strongest seasonal period.
- **Holiday campaigns** achieved the highest ROI, reaching nearly **200%**.
- **Black Friday V2** outperformed Black Friday V1 with an ROI of approximately **130%**.
- **New Year campaigns** generated the lowest revenue and delivered negative ROI.
- **Data Share Promo** attracted the most users but resulted in an ROI of approximately **-85%**.
- User activity was highest during the **middle of the week** and lowest during weekends.
- The **United States** was the top-performing market, generating approximately **\$991** in revenue with an average session duration of **13 minutes**.

